MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 29 – Oct 1 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. List the sample space and count outcomes to compute probabilities for equally likely outcomes, and apply the complement rule. (CLO 3) **2.** Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CLO 3)	e-Text: P.1 Probability Rules Interactive Lecture: P.1 Probability Rules Slides: P.1 Probability Rules	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ, σ, p, ρ vs. $\bar{x}$, s, $\hat{p}$, r). (CLO 5) **4.** Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CLO 5) **5.** Explain the standard error as the variability of a statistic and predict how sample size affects it. (CLO 5)	e-Text: 3.1 Sampling Distributions Interactive Lecture: 3.1 Sampling Distributions Slides: 3.1 Sampling Distributions	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	Online Homework: 3.1 Sampling Distributions (due Sun Oct 4, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Take-Home Quiz 3 (due Tue Sep 29, 11:59 PM) – covers 2.5 Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	Assignment Tool in Blackboard Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables.